Network Traffic Analysis using Wireshark

1. Introduction

This project focuses on analyzing network traffic using Wireshark to identify suspicious activities and potential cyber threats. The objective is to simulate a Security Operations Center (SOC) analyst role by investigating a real-world PCAP file.

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2. Objectives

- Capture and analyze network traffic
- Identify infected host
- Analyze DNS and HTTP traffic
- Detect phishing activity
- Identify Command & Control (C2) communication
- Extract system and user information

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3. Tools Used

- Wireshark

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4. Dataset Used

- PCAP file from Malware Traffic Analysis (Fake software download scenario)

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5. Analysis Performed

◆ 5.1 Protocol Analysis

- DNS traffic used for domain resolution
- HTTP traffic analyzed for web requests
- TCP used for communication tracking
- SMB/Kerberos used for authentication

◆ 5.2 Identification of Infected Host

- Internal network: `10.1.17.0/24`

- Suspicious traffic observed from: `10.1.17.215`

◆ 5.3 MAC Address Identification

- Extracted from packet details
- MAC Address: `00:d0:b7:26:4a:74`

◆ 5.4 Hostname Detection

- Found using NBNS/DHCP traffic
- Hostname: `DESKTOP-L8C5GSJ<00>`

◆ 5.5 User Account Identification

- Extracted from Kerberos/SMB authentication
- Username: `shutchenson`

◆ 5.6 Fake Domain Detection

- Identified using DNS queries
- Suspicious domain: `google-authenticator.burleson-appliance.net`
- Additional Suspicious domain: `authenticatoor.org`
- Reason: Not official Google domain

◆ 5.7 C2 Communication Detection

- External IP with repeated communication identified
- C2 IP: `45.125.66.32`

Indicators:

- Repeated connections
- Consistent traffic pattern
- Communication with infected host

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6. Anomalies Detected

- Unusual DNS queries

- Suspicious external communication
- Potential phishing activity
- Repeated beaconing traffic

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7. Conclusion

The analysis successfully identified an infected system and suspicious network behavior. Indicators of compromise (IOCs) such as fake domains and potential C2 communication were detected, demonstrating real-world threat analysis skills.

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8. Skills Gained

- Network Traffic Analysis
- Packet Inspection
- Threat Detection
- Incident Investigation
- Wireshark Proficiency